Joseph T. Race

jrace@iastate.edu
https://www.linkedin.com/in/joseph-race

Iowa State University
Department of Chemistry
2438 Pammel Drive
3266 Hach Hall
Ames, IA 50011

EDUCATION

Ph.D. in Chemistry from The Ohio State University — Columbus, OH
August 2019-October 2024

- Chemistry Graduate Program
- Dr. Patrick M. Woodward, Advisor
- Thesis: Understanding the Crystal Chemistry and Symmetry of Low-Dimensional Hybrid Organic–Inorganic Metal Halides

M.S. in Chemistry from The Ohio State University — Columbus, OH
August 2019-July 2021

- Chemistry Graduate Program
- Dr. Patrick M. Woodward, Advisor
- Ph.D. Candidacy Proposal: Exploring the Structure-Property Relationships of Perovskites and Delafossites

B.S. in Chemistry, Magna cum laude from Rochester Institute of Technology — Rochester, NY
August 2015-December 2018

- Chemistry Undergraduate Program
- Dr. Scott Williams, Advisor
- Research Description: Polycrystalline aluminum oxide and magnesium aluminum spinel films via a sol-gel route.

RESEARCH EXPERIENCE

Iowa State University and Ames National Laboratory — Ames, IA
November 2024-Present

- Postdoctoral Fellow in the Chemistry Department
- Dr. Kirill Kovnir, Mentor
- Synthesize and characterize candidate pnictides and chalcogenides for the absorbing layer of single and tandem junction solar cells. Working with a multidisciplinary team of theorists, film fabricators, and instrumentalists across several institutions.

The Ohio State University — Columbus, OH
May 2019-October 2024

- Graduate Research Associate
- Dr. Patrick M. Woodward, Advisor
- Synthesized and characterized novel organic-inorganic metal halides, focusing on crystal symmetries and the breaking of inversion symmetry.

Interprint — Pittsfield, MA
January-April 2019

- Ink Chemist
- Solved problems related to the rheology of inks and coatings used in décor printing.

Zero Valent NanoMetals — Rochester, NY
June-August 2016, 2017

- Research Scientist
- Developed and evaluated various metal nanoparticle ink formulations for inkjet printing.

Rochester Institute of Technology Rochester, NY

- Undergraduate Researcher January 2016-December 2018
- Dr. Scott Williams, Advisor
- Fabricated and characterized polycrystalline aluminum oxide and magnesium aluminum spinel films via a sol-gel route. Developed a gamma radiation sensitive dye system to be used in qualitative radiation dosimeters.

Rochester Institute of Technology Rochester, NY

- Undergraduate Researcher July 2015-December 2015
- Dr. Gerald Takacs, Advisor
- Investigated the surface modification of plastic films through VUV and UV photo-oxidation.

TEACHING EXPERIENCE

Iowa State University Ames, IA
Instructor of Record
Department of Chemistry

- 1st Semester General Chemistry Lecture (CHEM 1770), 170 students AU 2025

The Ohio State University Columbus, OH
Graduate Teaching Associate
Dept. of Chemistry & Biochemistry

- 1st Semester General Chemistry Lecture (CHEM 1210 Pilot Course supported by a grant from the Howard Hughes Medical Institute) AU 2023
- Graduate-level Solid State Chemistry (CHEM 7330) SP 2023
- 2nd Semester Honors General Chemistry Recitation and Lab (CHEM 1920H) SP 2022
- 1st Semester Honors General Chemistry Recitation and Lab (CHEM 1910H) AU 2021
- 2nd Semester General Chemistry Recitation and Lab (CHEM 1220) AU 2019-SP 2021

PUBLICATIONS

8) Yuan, Z.; …; **Race, J.T.**; …; Hautier, G. Long photoexcited carrier lifetime in a stable and earth-abundant zinc polyphosphide. *In revision*

7) **Race, J.T.**; Odeyemi, Y.; Woodward, P.M. CCDC 2472624: *CSD Communication*, 2025, 10.5517/ccdc.csd.cc2nzz2q.

6) **Race, J.T.**; Odeyemi, Y.; Woodward, P.M. CCDC 2472623: *CSD Communication*, 2025, 10.5517/ccdc.csd.cc2nzz1p.

5) **Race, J.T.**; Richtik, B.N.; Woodward, P.M. CCDC 2402915: *CSD Communication*, 2025, 10.5517/ccdc.csd.cc2lnfd3.

4) **Race, J.T.**; Richtik, B.N.; Woodward, P.M. CCDC 2402914: *CSD Communication*, 2025, DOI: 10.5517/ccdc.csd.cc2lnfc2.

3) Wang, H.; de la Torre, A.; **Race, J.T.**; Wang, Q.; Ruff, J.P.C.; Woodward, P.M.; Plumb, K.W.; Walker, D.; Xie, W. Pseudosymmetry in Tetragonal Perovskite $SrIrO_3$ synthesized under High Pressure. *ACS Appl. Electron. Mater.* **2024**, *6* (9), 6820–6825.

2) **Race, J.T.**; Foster, C.; Woodard, P.M. Chiral Templating in a Hybrid Chromium Chloride Hydrate. *Inorg. Chem*. **2024**, *63* (29), 13700–13706.

1) **Race, J.T.**; Liu, T.; Woodward, P.M. Structure Directing Forces in Hybrid Layered Double Perovskites Containing Aromatic Organic Cations. *Cryst. Growth Des.* **2024,** *24* (3), 1367–1379.

AWARDS & HONORS

Best Poster Award at the 53rd IEEE Photovoltaics Specialists Conference June 2025
IEEE

Undergraduate Research Award in Inorganic Chemistry SP 2019
Rochester Institute of Technology

School of Chemistry and Material Science Academic Research Scholar AU 2018
Rochester Institute of Technology

PRESENTATIONS

Race, J.T.; Coban, Y.; Rasmussen, G.; Hautier, G.; Kovnir, K. Prediction and Synthesis of Quaternary Metal Phosphides/File of Synthesis (FoS): A Universal File Format for Reporting Materials Synthesis. Poster presentation at the Solid State Chemistry Gordon Research Seminar and Conference, Colby-Sawyer College, New London, NH, July 2026.

Race, J.T.; Yuan, Z.; Goswami, S.; Viswanathan, G.; Quadir, S.; Bauers, S.; Liu, J.; Hautier, G.; Kovnir, K. Screening and Synthesis of Metal Phosphides as Novel Materials for Thin–Film Solar Absorbers. Poster presentation at the North American Solid State Chemistry Conference, Ames, IA, July 2025.

Race, J.T.; Yuan, Z.; Goswami, S.; Viswanathan, G.; Quadir, S.; Bauers, S.; Liu, J.; Hautier, G.; Kovnir, K. Screening and Synthesis of Metal Phosphides as Novel Materials for Thin–Film Solar Absorbers. Poster presentation at the IEEE Photovoltaics Specialists Conference, Montreal, QC, June 2025.

Race, J.T.; Woodward, P.M. Chiral Templating in a Hybrid Chromium Chloride Hydrate. Poster presentation at the Solid State Chemistry Gordon Research Seminar and Conference, Colby-Sawyer College, New London, NH, July 2024.

Race, J.T.; Liu, T.; Woodward, P.M. Structure directing forces in cation ordered hybrid layered perovskites. Poster presentation at the North American Solid State Chemistry Conference, Southern Alberta Institute of Technology, Calgary, AB, August 2023.

Race, J.T.; Woodward, P.M. Structure directing forces in hybrid layered perovskites: The search for new ferroelectric materials. Oral presentation during the Inorganic Chemistry Departmental Seminar at The Ohio State University, Columbus, OH, October 2022.

Race, J.T.; Liu, T.; Woodward, P.M. Crystal structure solutions of cation ordered hybrid layered perovskites. Poster presentation at the Pittsburgh Diffraction Conference, Lemont, IL, October 2022.

Race, J.T.; Woodward, P.M.; Clark, T.M. Supporting the inclusion of solid state chemistry in introductory courses. Oral presentation at the Biennial Conference on Chemical Education, West Lafayette, IN, August 2022.

Race, J.T.; Woodward, P.M. Lead-free layered hybrid perovskite ferroelectrics. Poster presentation at the Electronic Materials Conference, Columbus, OH, June 2022.

Race, J.T.; Woodward, P.M. Lead-free layered hybrid perovskite ferroelectrics. Poster presentation at the Hayes Graduate Research Forum, Columbus, OH, March 2022.

Race, J.T.; Woodward, P.M. Lead-free layered hybrid perovskite ferroelectrics. Poster presentation at Ohio State University Student Inorganic Poster Session, Columbus, OH, December 2021.

Race, J.T.; Williams, S. Absorption Study of Cu(II) Coordination Complexes with Tridentate Schiff Base Ligands. Poster Presentation at the 2018 MRS Fall Meeting & Exhibit, Boston, MA, November 2018.

SERVICE & LEADERSHIP ACTIVITIES

- DOE SCGSR Program Reviewer — SU 2025-Present
- Scouting America Chemistry Merit Badge Counselor — March 2025
- State Science Technology Fair of Iowa Judge — April 2025
- Iowa Regional Science Bowl Volunteer — February 2025
- Gordon Research Seminar on Solid State Chemistry 2026 Co-Chair — August 2024-July 2026
- Secretary of Joint Safety Team at OSU — SU 2022-SP 2023
- Scientific Thinkers member at OSU (Science Outreach for K-5 Students) — AU 2021-AU 2024
- Joint Safety Team member at OSU (Student-led research safety organization) — SU 2021-AU 2024
- ChemTALKS member at OSU (Organized the annual Mack Lecture) — SP 2020-AU 2024

PROFESSIONAL DEVELOPMENT & MEMBERSHIPS

- American Chemical Society — February 2026-Present
- Preparing Future Faculty Program Member at ISU — AU 2025-Present
- ISU Grantsmanship Microcredential: Finding Grants — September 2025
- IEEE Member — April 2025-SP 2026
- Postdoctoral Fellow Peer Speaking Practice Group at ISU — February 2025-Present
- Center for the Integration of Research, Teaching, and Learning (CIRTL) at OSU — AU 2021-SP 2022
- Grant Writing Seminar (HDFS 8862) at OSU — AU 2021
- RIT Chapter of the American Chemical Society — August 2017-December 2018

STUDENTS MENTORED

- Olivia Aguilar (Undergraduate student)- Iowa State University — SP 2026-Present
  - Research Project: Synthesis of Novel Metal Phosphides for Energy Generation
- Grason Rasmussen (Undergraduate student)- Iowa State University — SU 2025-SP 2026
  - Research Project: Synthesis of Novel Metal Phosphides for Energy Generation
- Jailin Bonkiye (Undergraduate student)- The Ohio State University — SP 2022-AU 2022
  - Research Project: Exploratory Synthesis of Hybrid Bismuth Halides

PROPOSALS WRITTEN

- ISU Postdoc Professional Development Grant- Funded — May 2026
- ISU First-Year Honors Mentor Program Mentor Grant- Funded — February 2026
- Advanced Photon Source General User Proposal (Run Cycle 2026-1, #1018038)- Awarded 3 shifts at Beamline 11ID-C — December 2025
- Ames Laboratory Directed Research and Development FY2026 Proposal- Not Funded — November 2025
- 2025 Arnold O Beckman Postdoctoral Fellowship in Chemical Instrumentation- Not Funded — AU 2025
- OSU College of Arts and Sciences Graduate Career Accelerator Award- Funded — SU 2023

REFERENCES

Kirill Kovnir, Ph.D.
Professor
Dept. of Chemistry
Iowa State University
3122 Hach Hall
2438 Pammel Dr
Ames, IA 50011
(515) 294-3111
kovnir@iastate.edu

Patrick M. Woodward, Ph.D.
Professor
Dept. of Chemistry & Biochemistry
The Ohio State University
257 CBEC Building
151 W Woodruff Ave
Columbus, OH 43210
(614) 688-8274
woodward.55@osu.edu

Ted Clark, Ph.D.
Professor of Teaching
Dept. of Chemistry & Biochemistry
The Ohio State University
120D Celeste Laboratory
120 W 18th Ave
Columbus, OH 43210
clark.789@osu.edu